

TUNKU ABDUL RAHMAN UNIVERSITY OF MANAGEMENT AND TECHNOLOGY

Group Contract Form

Faculty	:	Faculty of Computing and Information Technology
Programme	:	Bachelor in Information Technology (Honours) (Information Security)
Year of Study	:	3
Semester	:	202605
Course Code and Title	:	BMIT3173 Integrative Programming
Group Assignment Title	:	NutriShare: Surplus Food Redistribution Platform
Name of Tutor	:	Malarvili a/p Nallayan
Tutorial Group	:	4

Student Name & ID	Role(s)/Task(s) Assigned	Signature & Date
Liew Yi Ler (25WMR09747)	Module 1: Surplus Food Donation Publishing & Notification Management - Implemented surplus donation publishing pipeline, multi-criteria catalogue search, Leaflet.js geolocation visualizer, and RFC 4180 CSV export. Design Pattern: <i>Observer Pattern</i> (decoupled asynchronous notification dispatching to verified NGOs via background queue workers). Security Mitigations: CSRF protection via Synchronizer Token Pattern (<code>@csrf</code>) and Stored XSS mitigation via Context-Aware HTML Entity Escaping (<code>htmlspecialchars</code>). Web Services: Exposed <code>GET /api/donations/active</code> (IFA compliant); Consumed Module 2's <code>POST /api/user/verify-ngo</code>.	<i>Liew</i> 06/09/2026
Cheon Jie Han (25WMR09703)	Module 2: NGO Verification & Peer Trust Rating System Implemented statutory document verification and administrative audit queue, two-way idempotent peer trust reviews, and OTP password recovery.	<i>Cheon</i> 06/09/2026

	• Design Pattern: <i>Factory Method Pattern</i> (dynamic user onboarding and state encapsulation for Donors, NGOs, and Admins). • Security Mitigations: Broken Access Control mitigation via RBAC Middleware & FormRequest guards; OTP Brute-force mitigation via sliding expiry windows and IP rate limiting (<code>throttle:6,1</code>). • Web Services: Exposed <code>POST /api/user/verify-ngo</code> (IFA compliant); Consumed Module 3's <code>GET /api/claim/details</code>.	
Hiew Li Wei (25WMR09728)	Module 3: Claims & Logistics Distribution • Implemented claim reservation lifecycle, fleet transport vehicle dispatch, digital collection handover receipts (<code>REC-NUTRI-YYYYMMDD-XXX</code>), and SDG 2 distribution logging. • Design Pattern: <i>State Pattern</i> (governing claim lifecycle transitions: <i>Pending</i> → <i>Approved</i> → <i>Collected</i> while enforcing logistics invariants). • Security Mitigations: BOLA/IDOR mitigation via Policy-driven authorisation checks (<code>ClaimPolicy</code>); Parameter tampering & race condition mitigation via pessimistic row locking (<code>lockForUpdate()</code>). • Web Services: Exposed <code>GET /api/claim/details</code> (IFA compliant); Consumed Module 4's <code>POST /api/inventory/food-safety-check</code>.	<i>Hiew</i> 06/09/2026
Wong Men Jing (25WMR09788)	Module 4: Inventory & Food Safety Compliance • Implemented multi-location warehouse storage management, perishable food item intake, Many-to-Many allergen risk tagging, and RFC 4180 facility capacity audit exports. • Design Pattern: <i>Strategy Pattern</i> (runtime interchangeable notification delivery channels: <code>EmailStrategy</code> vs. <code>SMSStrategy</code>). • Security Mitigations: Log Injection (CRLF) mitigation via input stripping mutators (<code>SecurityHelper</code>); Parameter tampering mitigation on action links via HMAC SHA-256 signed URLs (<code>URL::signedRoute</code>).	<i>Wong</i> 06/09/2026

	• Web Services: Exposed GET /api/inventory/status & POST /api/inventory/food-safety-check; Consumed Module 1's GET /api/donations/active.	
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